Writing word equations

When carbon is heated it reacts with oxygen to form the product carbon dioxide. The two reactants, carbon and oxygen, are elements that form the compound carbon dioxide. This chemical reaction can be shown in the following **word equation**.

$$\text{carbon} + \text{oxygen} \xrightarrow{\text{heat}} \text{carbon dioxide}$$

We write 'heat' above the arrow to show that heat is needed for the reaction to occur.

☞ **1** For each of the following reactions:

- underline the reactants in red and the products in blue.
- write a word equation showing the reaction.

a When you mix the elements sulphur and oxygen together and heat them, the compound sulfur dioxide is produced.

_____

b Magnesium reacts with bromine to produce magnesium bromide.

_____

c When zinc is placed in hydrochloric acid it reacts, producing bubbles of hydrogen gas and a new substance called zinc chloride.

_____

d When iron reacts with oxygen it forms rust. The chemical name for rust is iron oxide.

_____

e Sodium reacts with chlorine to form sodium chloride.

_____

2 Use the information in the following word equation to complete the passage below.

$$\text{copper oxide} + \text{sulfuric acid} \longrightarrow \text{copper sulfate} + \text{water}$$

When copper _____ is placed in _____ acid it undergoes a chemical reaction. Two new substances are produced: copper _____ and _____.

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